

ANALOG CMOS CIRCUIT DESIGN SS26 - PRACTICAL EXERCISE V

Two-Stage Operational Amplifier

From Circuit Specifications to Transistor Dimensions

Tutors

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Analog CMOS Circuit Design SS26

Practical Exercise V – Circuit Specifications

Group

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Feedback

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Points **Mark**

Date **Signature**

No review without this standard form!



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In *Analog CMOS Circuit Design – Practical Exercise II*, the frequency response of the two-stage operational amplifier shown in Figure 1 was analyzed. It can be modeled as a 2-pole 1-zero system, i.e.,

$$\frac{u_{out}}{u_{in}} = A(s) = \frac{A_{DC}}{\left(1 + \frac{s}{|p_1|}\right) \left(1 + \frac{s}{|p_2|}\right)} = \frac{g_{m1} g_{m6} R_1 R_2}{\left(1 + \frac{s}{|p_1|}\right) \left(1 + \frac{s}{|p_2|}\right)} \frac{\left(1 - \frac{s}{|z_1|}\right)}{s}$$

where:

$$z_1 = \frac{g_{m6}}{C_M}, \quad p_1 \approx -\frac{1}{R_1 R_2 g_{m6} C_M}, \quad \text{and} \quad p_2 \approx -\frac{g_{m6}}{C_2 \left(1 + \frac{C_1}{C_M}\right)}.$$

In order to achieve an appropriate settling behavior in a closed-loop configuration, you have proven that $|p_2|$ must be placed higher than $2.2\omega_u$ (unity gain-bandwidth) while it was assumed that $z_1 > 10\omega_u$. Thus, a phase margin of at least 60° is achieved.

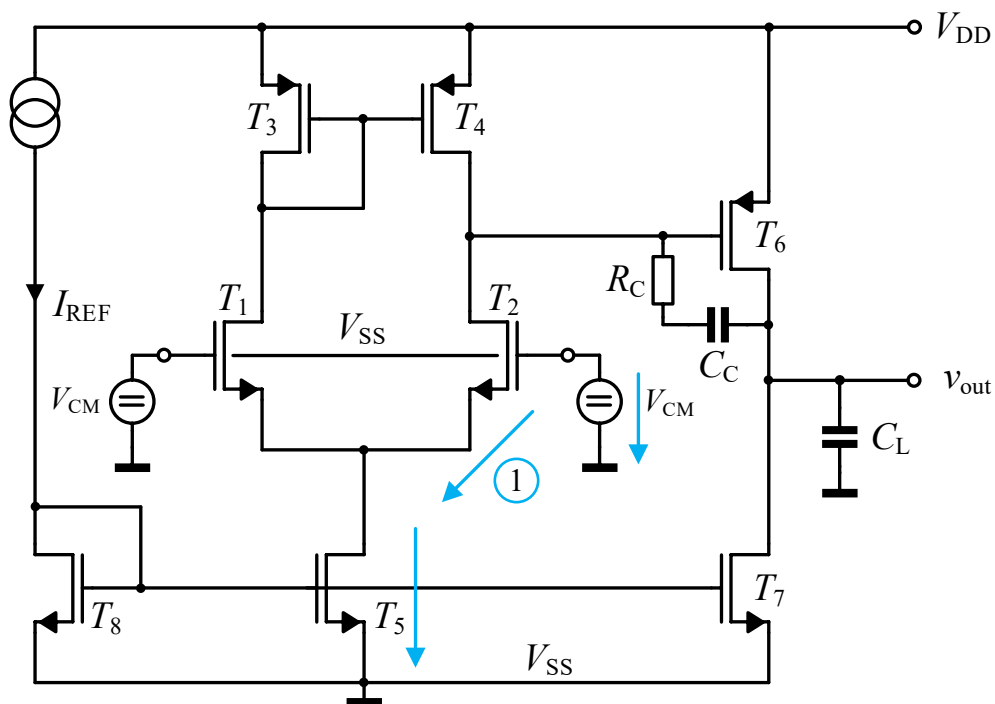


Figure 1: Two-stage Operational Amplifier



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This knowledge will be used in this practical exercise in order to design a two-stage operational amplifier (opamp) designated to drive a capacitive load C_L . The opamp will be implemented in a $0.35\mu\text{m}$ technology provided by *XFAB – Mixed Signal Foundry Experts*.

OPAMP Specifications

Load capacitance C_L	3pF
DC-Gain A_{DC}	$> 75\text{dB}$
Unity gain frequency ω_u	$> 2\pi \cdot 20\text{MHz}$
Phase margin PM	$\geq 60^\circ$
Positive and negative slew rate $ SR $	$> 15\text{V}/\mu\text{s}$
Input common mode range ICMR	$V_{CM,\min} = V_{SS} + 1\text{V} \leq \text{ICMR} \leq V_{CM,\max} = V_{DD} - 1\text{V}$
Input referred offset voltage V_{os}	Nominal value $ V_{os} \leq 200\mu\text{V}$ Standard deviation $ \sigma_{V_{os}} \leq 1.33 \text{ mV}$: $-3\sigma_{V_{os}} = -4\text{mV} \leq V_{os} \leq 4\text{mV} = +3\sigma_{V_{os}}$
Reference Current I_{ref}	$10\mu\text{A}$
Nominal supply voltage	$V_{DD} = 3.3\text{V}, V_{SS} = 0\text{V}$

General Notes:

- Transistors in analog designs rarely exhibit minimum length!
Rule of thumb:
Choose **3x minimum length** ($1\mu\text{m}$) for transistors which should provide gain!
Choose **6x minimum length** ($2\mu\text{m}$) for current mirrors!
- Body-Effect:** The bulk of any n-MOS transistor must be tied to V_{SS} . Due to the defined input common mode range (see above), the gate-source voltage of the input transistors T_1 and T_2 may thus be increased by 100 mV to 320 mV, depending on the bulk-source voltage.
- To keep all transistors in saturation, the minimum drain to source voltage should be

$$V_{DS} \geq 2 / \frac{g_m}{I_D} \quad (1)$$



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In addition, XFAB provided you with the following information for the n-MOS and p-MOS transistors of the 0.35 μm technology:

- Threshold voltage matching factors

$$A_{VT0n} = 6.18\text{mV}\cdot\mu\text{m} \quad A_{VT0p} = 4.632\text{mV}\cdot\mu\text{m}$$

Channel-length modulation modelled by $(1+\lambda V_{DS})$ as well as distance effects on the threshold- and β -mismatch can be neglected, i.e., $S_{VTn} = S_{VTp} = A_{\beta n} = A_{\beta p} = 0$.

T1: First Stage - Differential Amplifier

- Derive **C_M in terms of C_2** from the conditions that $z_1 > 10 \omega_u$ and $|p_2| > 2.2 \omega_u$.
Then **calculate the value of C_M** , while assuming, that C_L is the dominant contributor to C_2 . Hint: $C_1 / C_M \approx 0.3$.
- Derive the **minimum value for I_5** from the slew rate requirement, noting, that $I_5 = C_M \text{ SR}$ (the equation will be explained during the discussion of the practical exercise).
- Derive the required transconductance $g_{m1,2}$** from ω_u . The transistors minimum g_m/I_D -ratio is obtained by dividing $g_{m1,2}$ by the respective drain current.
Use the Matlab script to extract the normalized drain current of the transistor (e.g., $I/W [\text{A/m}] @ g_m/I_D$) and **calculate the corresponding width of the transistor**.
Hint: Which currents flow through T_1 and T_2 ?
- Derive W_5/L_5** from the specification of $V_{CM,\min}$ and **calculate the width W_5** .
Hints: Derive an equation for $V_{CM,\min}$ as a function of V_{DS5} and V_{GS} , by applying Kirchhoff's voltage law to the mesh 1 in Figure 1. To consider the body effect of the input transistors T_1 and T_2 take an increased threshold voltage of the input devices into account. Therefore extract V_{GS1} with the help of the g_m/I_D -scripts and add the margin given in the *general notes* to the resulting V_{GS} .
With this result, you can then calculate the minimum V_{DS5} and derive $(g_m/I_D)_5$ using equation 1. Using the normalized transistor current from the g_m/I_D -script you can then calculate the width of transistor T_5 .
- Derive $W_{3,4}/L_{3,4}$ and calculate their widths** from the specification of $V_{CM,\max}$, while taking the body effect of T_1 and T_2 into account. (*Task 1, e) continuous next page*)



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Hints: T_1 must remain in saturation and $V_{D1} = V_{D3} = V_{G3}$. With the resulting V_{G3} , a boundary for the $(g_m/I_D)_3$ ratio is obtained. The required width of the load transistors $W_{3,4}$ is then calculated by using the normalized transistor current (from the g_m/I_D scripts).

T2: Second-Stage – Inverter with Current Source Load

- a) **Derive and calculate the output stage current I_6** from the relation of pole p_2 and the phase margin PM.

Hint 1: First, derive the transconductance g_{m6} in terms of g_{m1} . In a second step determine the required $(g_m/I_D)_6$ ratio from the current mirror load and calculate I_6 .

Hint 2: Keep the *systematic offset caused by a “bad” design* in mind, e.g., $V_{DS,3} = V_{DS,4}$.

- What is the consequence regarding the g_m/I_D ratios of T_4 and T_6 ?
 - Which transistor determines $V_{DS,4}$, if the opamp is configured as a buffer, i.e., in unity gain configuration?
- b) **Derive W_6/L_6** from I_6 and the g_m/I_D scripts and **calculate the width of the transistor**.
- c) **Derive W_7/L_7** from I_7/I_5 and **calculate the width of the transistor**.

T3: Input referred offset voltage

- a) **Identify** the two transistor pairs which determine the input referred offset voltage of the operational amplifier! **Shortly explain** why these transistors define the offset voltage!
- b) **Calculate the standard deviation of the input referred offset voltage** based on the transistor dimensions from Task 1 and Task 2, the result from Task 3 a) and the threshold mismatch parameter A_{VTOn} and A_{VTOp} .
- c) **Redesign these transistors** in order to meet the specification for the input referred offset voltage, i.e., recalculate their dimensions!

